

Table 1 – Q2 values on validation data (dependency on number of latent variables)									
Latent variables	1	2	3	4	5	6	7	8	9
OS1, PCR, target 17, w 5, fold 3, all sensors	0.5769	0.6041	0.6041	0.6042	0.6043	0.6043	0.6042	0.6041	0.6040
OS1, PLS, target 17, w 5, fold 3, all sensors	0.5879	0.6043	0.6043	0.6043	0.6021	0.6014	0.6008	0.6007	0.6004
OS3, PCR, target 21, w 5, fold 1, all sensors	0.1685	0.7706	0.8103	0.8246	0.8328	0.8330	0.8332	0.8333	0.8333
OS3, PLS, target 21, w 5, fold 1, all sensors	0.7374	0.7996	0.8297	0.8316	0.8322	0.8328	0.8343	0.8351	0.8350

Table 2 – cumulative explained variance in training data (dependency on number of latent variables) [%]									
Latent variables	1	2	3	4	5	6	7	8	9
OS1, PCR, target 17, w 5, fold 3, all sensors	63.45	76.71	77.74	78.66	79.43	80.16	80.84	81.43	82.01
OS1, PLS, target 17, w 5, fold 3, all sensors	63.39	76.71	77.48	78.18	78.58	79.03	79.45	79.99	80.62
OS3, PCR, target 21, w 5, fold 1, all sensors	49.96	73.19	78.64	83.41	85.04	85.87	86.65	87.36	88.00
OS3, PLS, target 21, w 5, fold 1, all sensors	35.10	73.08	78.25	80.36	84.60	85.52	85.98	87.07	87.07

Table 1 – Operational setting 1 (target Sensor 17), PCR					
Hyperparameters setup	Fold	Training (RMSE, R2)	Calibration (RMSE, Q2)	Validation (RMSE, Q2)	Testing (RMSE, Q2)
window size = 5 k = 3 all sensors	1	0.9478, 0.6099	0.9470, 0.6105		
	2	0.9459, 0.6160	0.9508, 0.5996		
	3	0.9486, 0.6037	0.9454, 0.6232	0.9716, 0.6041	
window size = 5 k = 5 all sensors	1	0.9472, 0.6105	0.9472, 0.6104		
	2	0.9455, 0.6163	0.9505, 0.5999		
	3	0.9483, 0.6040	0.9449, 0.6236	0.9713, 0.6043	0.9643, 0.5899
window size = 10 k = 3 all sensors	1	0.9463, 0.5973	0.9461, 0.5957		
	2	0.9451, 0.6018	0.9484, 0.5881		
	3	0.9467, 0.5905	0.9456, 0.6092	0.9715, 0.5865	
window size = 10 k = 5 all sensors	1	0.9462, 0.5974	0.9462, 0.5956		
	2	0.9450, 0.6019	0.9483, 0.5881		
	3	0.9465, 0.5907	0.9457, 0.6091	0.9715, 0.5866	
window size = 5 k = 3 S 3, 11	1	0.9692, 0.5922	0.9670, 0.5940		
	2	0.9666, 0.5990	0.9719, 0.5817		
	3	0.9689, 0.5866	0.9674, 0.6055	1.001, 0.5800	
window size = 5 k = 5 S 3, 11	1	0.9690, 0.5924	0.9672, 0.5938		
	2	0.9666, 0.5990	0.9720, 0.5816		
	3	0.9688, 0.5867	0.9674, 0.6056	1.001, 0.5799	
window size = 10 k = 3 S 3, 11	1	0.9608, 0.5849	0.9568, 0.5865		
	2	0.9584, 0.5906	0.9616, 0.5766		
	3	0.9586, 0.5802	0.9614, 0.5962	0.9901, 0.5709	
window size = 10 k = 5 S 3, 11	1	0.9608, 0.5850	0.9568, 0.5866		
	2	0.9583, 0.5907	0.9619, 0.5763		
	3	0.9586, 0.5803	0.9615, 0.5961	0.9902, 0.5707	

Table 2 – Operational setting 1 (target Sensor 17), PLS					
Hyperparameters setup	Fold	Training (RMSE, R2)	Calibration (RMSE, Q2)	Validation (RMSE, Q2)	Testing (RMSE, Q2)
window size = 5 k = 3 all sensors	1	0.9470, 0.6106	0.0471, 0.6105		
	2	0.9482, 0.6141	0.9538, 0.5971		
	3	0.9478, 0.6044	0.9451, 0.6234	0.9714, 0.6043	0.9641, 0.5901
window size = 5 k = 5 all sensors	1	0.9453, 0.6120	0.9480, 0.6097		
	2	0.9462, 0.6157	0.9553, 0.5958		
	3	0.9460, 0.6059	0.9464, 0.6224	0.9740, 0.6021	
window size = 10 k = 3 all sensors	1	0.9462, 0.5974	0.9466, 0.5953		
	2	0.9503, 0.5973	0.9548, 0.5824		
	3	0.9462, 0.5910	0.9465, 0.6085	0.9725, 0.5857	
window size = 10 k = 5 all sensors	1	0.9444, 0.5989	0.9482, 0.5939		
	2	0.9484, 0.5989	0.9567, 0.5807		
	3	0.9445, 0.5924	0.9477, 0.6075	0.9744, 0.5841	
window size = 5 k = 3 S 3, 11	1	0.9673, 0.5938	0.9634, 0.5970		
	2	0.9668, 0.5988	0.9720, 0.5816		
	3	0.9656, 0.5894	0.9670, 0.6059	0.9988, 0.5817	
window size = 5 k = 5 S 3, 11	1	0.9670, 0.5941	0.9636, 0.5968		
	2	0.9666, 0.5990	0.9724, 0.5813		
	3	0.9654, 0.5896	0.9669, 0.6059	0.9991, 0.5815	
window size = 10 k = 3 S 3, 11	1	0.9608, 0.5850	0.9576, 0.5859		
	2	0.9624, 0.5872	0.9652, 0.5734		
	3	0.9580, 0.5808	0.9614, 0.5962	0.9896, 0.5713	
window size = 10 k = 5 S 3, 11	1	0.9603, 0.9582	0.9582, 0.5853		
	2	0.9619, 0.5876	0.9655, 0.5731		
	3	0.9578, 0.5810	0.9614, 0.5963	0.9900, 0.5709	

Table 3 – Operational setting 3 (target Sensor 21), PCR					
Hyperparameters setup	Fold	Training (RMSE, R2)	Calibration (RMSE, Q2)	Validation (RMSE, Q2)	Testing (RMSE, Q2)
window size = 5 k = 3 all sensors	1	0.0632, 0.8065	0.0631, 0.8266	0.0659, 0.8103	
	2	0.0631, 0.8186	0.0612, 0.8139		
	3	0.0622, 0.8197	0.0645, 0.8040		
window size = 5 k = 5 all sensors	1	0.0612, 0.8181	0.0612, 0.8328	0.0619, 0.8328	0.0608, 0.8332
	2	0.0615, 0.8279	0.0606, 0.8176		
	3	0.0608, 0.8274	0.0619, 0.8199		
window size = 10 k = 3 all sensors	1	0.0625, 0.8020	0.0622, 0.8238	0.0641, 0.8143	
	2	0.0626, 0.8141	0.0608, 0.8079		
	3	0.0612, 0.8175	0.0634, 0.8033		
window size = 10 k = 5 all sensors	1	0.0610, 0.8114	0.0609, 0.8313	0.0616, 0.8283	
	2	0.0612, 0.8219	0.0604, 0.8103		
	3	0.0606, 0.8208	0.0617, 0.8277		
window size = 5 k = 3 S 7, 20	1	0.0654, 0.7919	0.0657, 0.8122	0.0662, 0.8084	
	2	0.0659, 0.8025	0.0648, 0.7914		
	3	0.0652, 0.8021	0.0662, 0.7935		
window size = 5 k = 5 S 7, 20	1	0.0652, 0.7936	0.0657, 0.8130	0.0660, 0.8098	
	2	0.0657, 0.8034	0.0645, 0.7935		
	3	0.0650, 0.8037	0.0661, 0.7944		
window size = 10 k = 3 S 7, 20	1	0.0634, 0.7964	0.0634, 0.8174	0.0640, 0.8146	
	2	0.0637, 0.8071	0.0626, 0.7961		
	3	0.0630, 0.8068	0.0642, 0.7982		
window size = 10 k = 5 S 7, 20	1	0.0634, 0.7970	0.0634, 0.8176	0.0640, 0.8148	
	2	0.0637, 0.8074	0.0626, 0.7996		
	3	0.0630, 0.8073	0.0642, 0.7984		

Table 4 – Operational setting 3 (target Sensor 21), PLS					
Hyperparameters setup	Fold	Training (RMSE, R2)	Calibration (RMSE, Q2)	Validation (RMSE, Q2)	Testing (RMSE, Q2)
window size = 5 k = 3 all sensors	1	0.0618, 0.8134	0.0617, 0.8343	0.0624, 0.8297	
	2	0.0619, 0.8254	0.0607, 0.8170		
	3	0.0613, 0.8246	0.0629, 0.8139		
window size = 5 k = 5 all sensors	1	0.0712, 0.8177	0.0613, 0.8363	0.0620, 0.8322	0.0609, 0.8328
	2	0.0614, 0.8286	0.0605, 0.8183		
	3	0.0610, 0.8268	0.0620, 0.8192		
window size = 10 k = 3 all sensors	1	0.0617, 0.8071	0.0615, 0.8279	0.0623, 0.8246	
	2	0.0617, 0.8191	0.0605, 0.8096		
	3	0.0611, 0.8180	0.0627, 0.8077		
window size = 10 k = 5 all sensors	1	0.0612, 0.8100	0.0611, 0.8301	0.0619, 0.8269	
	2	0.0612, 0.8221	0.0604, 0.8104		
	3	0.0608, 0.8196	0.0619, 0.8127		
window size = 5 k = 3 S 7, 20	1	0.0656, 0.7908	0.0658, 0.8118	0.0666, 0.8077	
	2	0.0659, 0.8025	0.0648, 0.7915		
	3	0.0656, 0.7993	0.0666, 0.7914		
window size = 5 k = 5 S 7, 20	1	0.0653, 0.7930	0.0655, 0.8134	0.0660, 0.8096	
	2	0.0655, 0.8044	0.0644, 0.7940		
	3	0.0652, 0.8016	0.0663, 0.7932		
window size = 10 k = 3 S 7, 20	1	0.0636, 0.7948	0.0635, 0.8165	0.0642, 0.8136	
	2	0.0638, 0.8069	0.0627, 0.7958		
	3	0.0634, 0.8039	0.0646, 0.7959		
window size = 10 k = 5 S 7, 20	1	0.0635, 0.7953	0.0635, 0.8166	0.06415, 0.8138	
	2	0.0636, 0.8077	0.0626, 0.7966		
	3	0.0633, 0.8046	0.0645, 0.7963		